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# Intelligence Engineering

## Solving Real World Problems At Scale

Prof. Dr.-Ing. Mark Schutera | September 25, 2026 · *drifting stream*

LECTURE V

# Feature Engineering

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# This lecture in one slide

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## V Feature Engineering

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Learned Features Versus Engineered Features

Common Feature Engineering Operations

Data Leakage

Engineering Good Features

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*why this lecture matters, and what  
you can do after it*

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# Learned Features Versus Engineered Features

# Learned Features Versus Engineered Features

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**Learned features**

**Engineered features**

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2

# Common Feature Engineering Operations

# Handling Missing Values

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$P(y | x)$

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[1]

# Deletion

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**Definition: Deletion**

# Imputation

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$P(\hat{y} | x)$

Imputation

[1]

# Scaling

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$P(\hat{y} | x)$

[1]

# Discretization

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$P(y | x)$

Discretization

[1]

# Encoding Categorical Features

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$P(y | x)$

[1]

# Feature Crossing

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Feature crossing

# Discrete and Continuous Positional Embeddings

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**Discrete**

**Continuous**

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3

# Data Leakage

# Common Causes for Data Leakage

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$P(y|x)$

**Time-correlated splits**

**Scaling before splitting**

**Test split statistics**

**Data duplication**

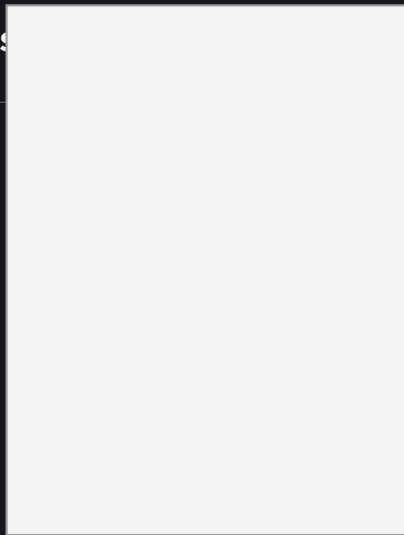
**Group leakage**

**Data generation process**

[1]

## Splitting time-correlated data randomly into

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$P(y | x)$

[1]

# Scaling before splitting

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Scaling before splitting

$P(y | x)$

[1]

# Filling in missing data with statistics from the test split

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$P(y | x)$

[1]

# Poor handling of data duplication before splitting

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**Definition: Data duplication**

# Group leakage

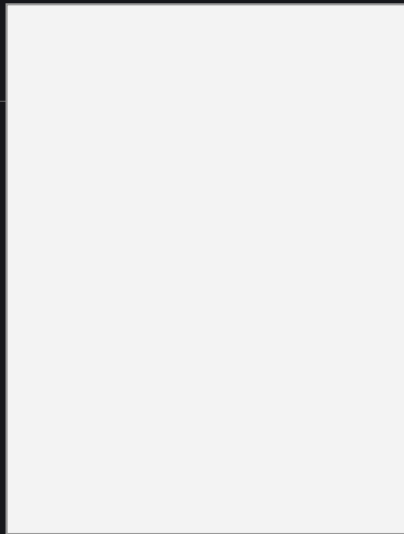
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$$P(\hat{y} | x)$$

$$[1]$$

# Leakage from data generation process

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$P(y | x)$

[1]

# Detecting Data Leakage

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$P(\hat{y} | x)$

[1]

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4

# Engineering Good Features

# Feature Importance

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| Feature | Importance |
|---------|------------|
|---------|------------|

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# Feature Generalization

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$P(\hat{y} | x)$

[1]

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# Summary

# Summary

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$P(y|x)$

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[1]

# References

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- 1 Chip Huyen. *Designing Machine Learning Systems: An Iterative Process for Production-Ready Applications*. Sebastopol, CA: O'Reilly Media, 2022.